



AI Data Assistant Chatbot — Project Statistics & Summary

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PROJECT STATISTICS

Metric	Value	Metric	Value
API Endpoints	8	Core Files	5
Technologies Used	15+	Cloud Platforms	2
Lines of Code	~550	Build Time	1 Day
Female Voice Options	3	GitHub Repos	2
File Types Supported	CSV, TXT	Deployment Platforms	Vercel + Render

FEATURES

Feature	Technology	Status
AI Chat with Conversation Memory	Claude AI (Anthropic)	■ Live
Voice Input (Microphone)	Web Speech API — Chrome built-in	■ Live
Voice Output (Female — Heera)	Speech Synthesis API	■ Live
CSV File Upload & Analysis	Pandas + Claude AI	■ Live

Text File Summarization	Claude AI	■ Live
Per-Message Summarize Button	Claude AI	■ Live
Natural Language to SQL	Claude AI + SQLAlchemy	■ Live
Clear Chat History	FastAPI + React State	■ Live

TECH STACK

Layer	Technologies
Frontend	React • TypeScript • Axios • Recharts • Web Speech API
Backend	Python 3.12 • FastAPI • Uvicorn • Pandas • NumPy
AI & Voice	Claude AI (claude-sonnet-4-6) • Anthropic SDK • Speech Synthesis API
Cloud	AWS (EC2, S3, RDS) • Microsoft Azure (AZ-900, AZ-204 Certified)
Hosting	Vercel (Frontend) • Render (Backend) — Both Free Tier
DevOps	Git • GitHub • CI/CD via Vercel Auto-Deploy on Push
Databases	PostgreSQL • MongoDB • MySQL • SQL

LIVE LINKS & RESOURCES

Resource	URL
Live Chat App	https://datamind-frontend-henna.vercel.app
Backend API Docs	https://datamind-backend-u9nm.onrender.com/docs
GitHub — Frontend	https://github.com/MarellaMrudula/datamind-frontend
GitHub — Backend	https://github.com/MarellaMrudula/datamind-backend
LinkedIn	https://www.linkedin.com/in/marella-mrudula

Get API Key	https://console.anthropic.com
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SKILLS DEMONSTRATED

Skill	Evidence in This Project
Python & FastAPI	Built REST API backend with 8 endpoints and full CORS configuration
React & TypeScript	Built full chat UI with hooks, state management, and file handling
AI / ML Integration	Integrated Claude AI with conversation memory and streaming
Data Engineering	CSV processing, ETL pipeline, Pandas data analysis
Cloud Deployment	Deployed on Vercel + Render; hands-on AWS and Azure certified
Voice Technology	Web Speech API (input) + Speech Synthesis API (female output)
DevOps & Git	GitHub repositories with CI/CD auto-deploy via Vercel
SQL & Databases	NLP to SQL conversion, PostgreSQL/MongoDB knowledge
REST API Design	8 endpoints with proper HTTP methods, CORS, and JSON responses
Full-Stack Dev	End-to-end: Python backend + React frontend + cloud hosting